Reconstruct Multiscale Features for Lightweight Small Object Detection in Remote Sensing Images

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***Abstract*—Most small objects are missed when object detection algorithms are transferred from natural images to remote sensing images. Constructing multiscale features has been proven to be an effective approach for detecting small objects. However, existing methods for multiscale features have two limitations: insufficient discriminative capability and sparse semantic-spatial information, which fail to fully leverage the potential of multiscale features. To overcome these limitations, we propose the multiscale feature reconstruction network (MRN), which introduces three novel modules during feature extraction, fusion, and enhancement: the composite multiscale feature extraction module (CEM), interlayer feature joint module (IJM), and spatial-semantic information cross module (SSM). First, CEM utilizes a multibranch structure to aggregate scale information. Dilated convolution and asymmetric convolution are extensively used in the branches, which expand the receptive field and capture information of rectangular instances, respectively. Second, the IJM leverages a gating mechanism to achieve pixel-level feature enhancement for feature maps at different hierarchical depths. Finally, the SSM alleviates high-level semantic feature information imbalance through dual-branch information interaction. Furthermore, to utilize the limited computational resources, we propose a lightweight version called MRN_lite. We evaluate MRN and MRN_lite on three existing public datasets: AITOD, Vehicle Detection in Aerial Imagery (VEDAI), and VisDrone2019. Extensive experiments demonstrate the effectiveness of our method. In comparison experiments, both versions of the model outperform the state of the art (SOTA). And MRN_lite has less than 50% of the FLOPs and parameters of MRN, which has comparable performance to the original version.**

***Index Terms*—Feature enhancement and fusion, lightweight network, multiscale features, remote sensing image, small object detection.**

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Fig. 1. Sparse pixel units of objects in remote sensing images.

I. Introduction

OBJECT detection technology has garnered significant attention since its inception, with numerous prominent works being proposed in succession, such as Faster RCNN [1], Cascade RCNN [2], SSD [3], YOLO [4], and DETR [5]. With the advancement of remote sensing technology, object detection technology holds broad application prospects in fields such as environmental protection [6], [7], urban planning [8], disaster monitoring [9], [10], [11], and military reconnaissance [12], [13]. However, when applying object detection algorithms from natural images to remote sensing images, the detection performance is often unsatisfactory, particularly for small-scale objects. For example, the DETR model achieves detection accuracies of 12.4% and 3.0% for medium-sized and small-sized objects, respectively, on the AITOD dataset. This is primarily due to the sparse pixel units of objects in remote sensing images, as shown in Fig. 1. Useful objects account for a very small proportion of the entire image, and tiny objects contain only a few pixels, making them difficult to distinguish even with the naked eye.

Sparse pixel units result in insufficient object information, and supplementing adequate feature representation from various scales is crucial for enhancing model performance. As shown in Fig. 2, remote sensing images have a high proportion of small objects, with pixels typically less than 32×32 [14], [15]. Such pixel scarcity leads to blurring of object texture and edge, making it challenging for models to accurately separate the objects from the background in complex and diverse scenes [16], [17], [18], [19], [20]. Existing methods primarily fall into three categories: data augmentation [21], [22], [23], image super-resolution [24], [25], [26], [27], and feature pyramid network (FPN) [28], [29], [30], [31]. Unlike the previous two methods, FPN does not impose a significant computational burden on the platform. FPN constructs a top-down architecture with lateral connections, constructing high-level semantic feature maps at all scales and demonstrating excellent performance in small object detection. AugFPN [29] added a residual connection to FPN, which reduces the disparity of semantic information at different scales and

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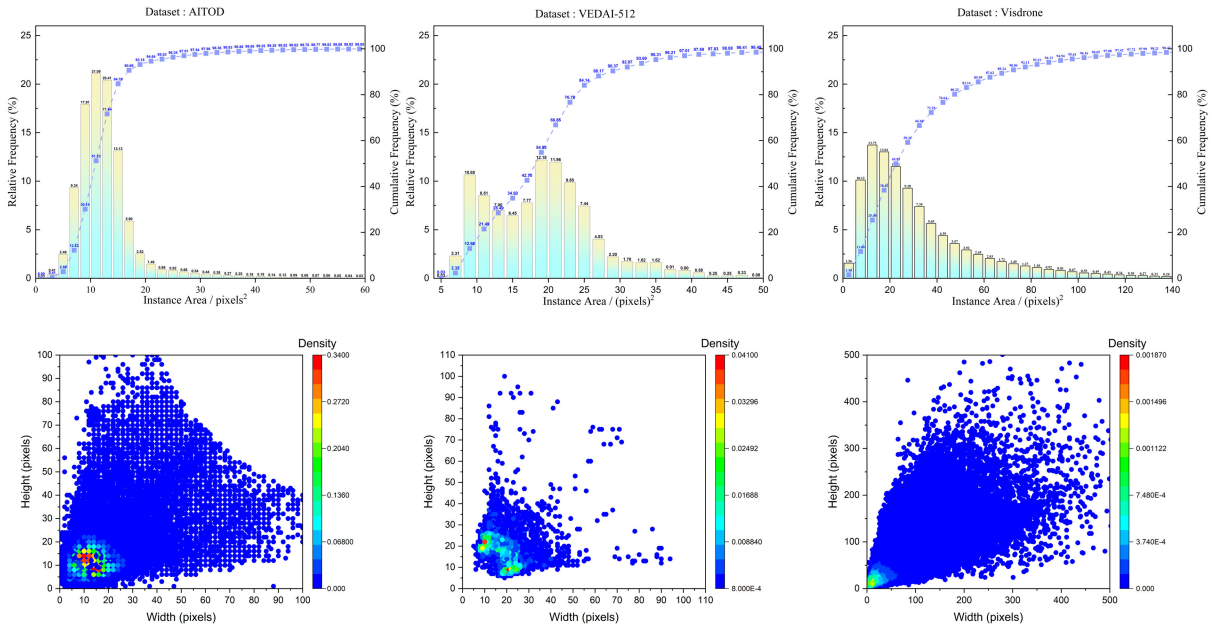


Fig. 2. Visualization of the instance size of the three datasets. Histogram of instance size distribution (first row) and scatter density plot of instance size (second row).

compensates for the missing details in high-level features. However, conventional approaches generally perform feature extraction, enhancement, and fusion separately at each scale. They then combine the processed features from different levels to form multiscale representations. This strategy produces features largely independent across scales, failing to exploit the interscale relationships. In contrast, the multiscale feature reconstruction network (MRN) proposes a novel strategy for constructing multiscale features. Rather than combining features in later stages, MRN extracts multiscale features at an early stage in the model. This design enables the active fusion and enhancement of these features throughout the network's forward propagation, yielding more effective representations.

MRN consists of three modules, each designed to optimize multiscale feature extraction, fusion, and enhancement, respectively. To extract multiscale features, the composite multiscale feature extraction module (CEM) introduces a multibranch structure with asymmetric and dilated convolution layers. Each branch is designed to capture and aggregate features with different receptive fields. Subsequently, the interlayer feature joint module (IJM) enhances the discriminative ability of multiscale features by extracting differences between adjacent features. Previous work has indicated that deeper networks can extract global semantic information [32], [33]. As network layers increase, features pass through multiple convolutional and pooling layers, resulting in a larger receptive field, but also causing a continuous loss of spatial information. Shallow features, with smaller receptive fields, can only describe spatial details such as textures and edges in images [34], [35]. This imbalance of information is particularly detrimental to small object detection. The spatial-semantic information cross module (SSM) facilitates the interaction between multiscale features to mitigate the loss of spatial information.

Furthermore, due to the massive data generated by remote sensing platforms, which takes a long time to transmit, the traditional ground detection mode is hard to meet the time efficiency requirements of rescue and disaster relief tasks. With the iterative upgrading of remote sensing manufacturing and chip technology, remote sensing data has achieved multidimensional breakthroughs in spatial, spectral, and temporal resolution, marking the remote sensing big data era. Currently, data acquisition capabilities have significantly outpaced information processing efficiency, leading to a severe imbalance between cognitive capacity and data growth [36]. Real-time object detection demands a transformation of the detection mode to achieve shorter time delays, and the traditional "transmission followed by detection" approach is gradually evolving into "detection followed by transmission." Thus, developing a lightweight model that could be deployed in the payload platform becomes an important research direction. On-orbit object detection technology can reduce data transmission volume and detection latency, thus emerging as one of the core elements in the construction of intelligent remote sensing satellite systems [37]. However, the deployment of on-orbit detection models is constrained by resources such as storage and computing power, and striking a balance between accuracy and cost constitutes the central theme of lightweight design.

Building upon the concepts discussed above, we propose a lightweight version called MRN_lite. In the lightweighting process, group convolutions and depth-wise convolutions with fewer parameters are extensively used to replace ordinary convolutions. In addition, we employ shared weights to reduce the number of convolutions. Thanks to the efficient utilization of parameters in the proposed network architecture, MRN_lite achieves performance comparable to that of the MRN model.

In short, we hope to propose an accurate and reliable lightweight model to detect small objects in real time. A rich feature representation can improve the model's ability to recognize and localize, so constructing a multiscale feature

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with stronger discrimination power is the focus of this work. Meanwhile, to address the challenges of the remote sensing big data era, we will lightweight the model so that it has the potential to be deployed on remote sensing platforms in the future.

The main contributions of this article are shown below.

1) In this article, we propose an MRN. The reconstructed features aggregate discriminative scale information for the recognition and localization of small objects. To alleviate latency issues caused by insufficient transmission bandwidth, this article also proposes a lightweight version of MRN_lite. It can be deployed on the platform to enable detection followed by transmission, reducing data transmission volume and detection latency.
2) The MRN network includes three novel modules: CEM, IJM, and SSM. CEM is capable of extracting features with different sampling coverage and feature information of asymmetric objects. IJM focuses on filtering useless information and enhancing the key information in the fusion process, while SSM can supplement the deeper features with more refined spatial information.
3) Extensive experiments on three representative small object datasets in remote sensing (AITOD, Vehicle Detection in Aerial Imagery (VEDAI), and VisDrone2019), which demonstrated the effectiveness of MRN and MRN_lite, with both outperforming state of the art (SOTA). Furthermore, detailed ablation experiments were conducted on each module to thoroughly investigate and explore the contributions of each component.

II. Related Works

A. Small Object Detection in Remote Sensing

Object detection is one of the fundamental tasks in computer vision, aimed at identifying objects and obtaining their locations. Detection algorithms are typically categorized into two types: two-stage detectors (RCNN [38], Faster RCNN [1], and Cascade RCNN [2]) and single-stage detectors (YOLO series [4], [39], [40], [41], SSD [3], and RetinaNet [42]). Two-stage detectors offer high localization and recognition accuracy but come with higher computational costs. Single-stage detectors have been iterated and innovated, matching or even surpassing the accuracy of two-stage algorithms while maintaining speed. Among single-stage algorithms, the YOLO model is widely adopted due to its high flexibility and adaptability.

Compared to natural images, objects in remote sensing images have a higher proportion, smaller sizes, and less information. Insufficient information can deteriorate the feature representation that the model can learn. To solve this dilemma, the mainstream methods in remote sensing are categorized into three approaches: data augmentation, image super-resolution, and FPNs.

1) Data Augmentation: Data augmentation methods for small objects in remote sensing mainly include cropping, splicing, and scaling methods, which enrich the number of instances. The Mosaic data augmentation technique [21] splices multiple images together, which increases the diversity and complexity of the training data and thus reduces the risk of overfitting. RRNet [22] utilizes the copy-paste strategy to increase the instances; they propose an adaptive enhancement strategy named AdaResampled that can effectively eliminate two mismatching phenomena in the copy-paste process: background mismatching and scale mismatch. However, the copy-paste method is less efficient when applied to large datasets. Divided image based on the clustering center division of the true value of the object, removing invalid background information before data enhancement, and reducing the time cost of training [23]. Cropping increases the proportion of truncated objects and introduces scale variation in objects, thereby enabling the network to better learn partial features [43]. Data augmentation methods have limited effectiveness and significantly increase the computational burden on large datasets.

2) Image Super-Resolution: GAN [24] has the powerful image generation capability, and small objects can be augmented by estimating a high-resolution (HR) image from its low-resolution (LR) counterpart. Li et al. [25], Ledig et al. [26], Denton et al. [27], and Ji et al. [44] utilize GAN networks to strengthen the weak feature representation of tiny objects and reduce the representation difference between object sizes. Subsequently, super-resolution technology based on GAN networks began to be applied to small objects in remote sensing. EESRGAN [45] filters edge details during the GAN image reconstruction process, effectively avoiding the loss of high-frequency edge information of objects. HRFG [46] extracted real HR feature supervision signals, which guide the generator network to extract HR features from LR images directly. The super-resolution technique provides a practical and effective path to improve the detection of tiny objects. However, the additional cost of computational resources brought by super-resolution techniques cannot be ignored, and it brings new problems, such as pseudo-textures and artifacts.

3) Feature Pyramid Networks: The FPNs [28] with rich layer features can address the challenge of detecting small objects. Multiple-layer features are combined to construct multiscale features to predict objects at the corresponding scale. Liu et al. [30] and Li et al. [31] found that the use of multiscale training strategies can effectively achieve the average accuracy of small and medium-sized instances. SDP [47] and MS-CNN [48] output suitable features for various-sized objects so that different receptive fields match objects of corresponding scales. Recursive feature pyramid (RFP) [49] uses additional feedback connections to bring features directly from the detection head back into the bottom-up backbone layer, which significantly improves the performance of object detection. To achieve multiscale object detection from complex optical remote sensing images, a densely connected FPN provides high-quality information for multiscale appearance object detection tasks through advanced multiscale semantic feature maps [50]. For remote sensing images, the global-to-local scale-aware detection network (GLSANet) [51] obtains different perspective content by rotating the high-level semantic features of FPN. In this way, the interaction between perspectives can be used to further strengthen global semantic information.

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We will reconstruct the multiscale features of FPN, which increase scale information about objects and assist in model localization and recognition.

B. Multiscale Feature Reconstruction

Reconstruction at the feature level is mainly divided into two types. Reconstruct features through feature layer decomposition or channel pruning to accelerate the computation of convolutional neural networks [52], [53]. The other methods involve reconstructing features with strong representational capability by separating strong and weak feature information or reweighting features across channels and spatial dimensions [54], [55]. Like the last method, MRN aims to enhance the representational power of multiscale features. The reconstruction involves three aspects: feature extraction, fusion, and enhancement. We will also introduce related work from these three aspects.

By extracting rich feature representations and optimizing the feature fusion strategy, the model can obtain higher quality features and improve the detection accuracy. AugFPN [29] extracts ratio-invariant contextual information to minimize the information loss in feature fusion. FFCA-YOLO [56] is proposed to extract semantic information and fuse multiscale features using a reweighting strategy. MENet [57] assigns weights to the features from the spatial direction as well as the channel direction, respectively, and solves the problem of fusion of neighboring stages of feature maps. G-FRNet [58] is used to modulate the feature information passed through the hopping connections, which removes ambiguous information from the underlying encoder network during fusion with the decoder network. Auto-FPN [59] and NAS-FPN [60] use a neural network to learn feature pyramid architecture for cross-scale connections to realize cross-scale fusion functions. BiFPN [61] repeats the top-down as well as bottom-up bidirectional paths for feature fusion.

As the network deepens, the imbalance between spatial and semantic information increases. LR deep feature maps, rich in semantic information, lack spatial detail and consequently suffer in object localization. Conversely, HR shallow feature maps contain abundant spatial details and limited semantic information, resulting in underperformance in object recognition. Combining deep features with shallow features, the poor detection performance caused by information imbalance can be effectively avoided. To minimize the impact of information imbalance, QueryDet [62] first predicted rough locations of tiny objects on an LR feature map, and these rough locations guided a HR feature map to compute exact locations. Inspired by [32] and [33], the short paths from shallow features to deep features can avoid spatial information loss. Dense convolutional networks (DenseNet) [63] input previous feature maps to all the subsequent layers. IPG-Net [35] proposed the IPG transform module for extracting detailed spatial information from shallow features and addressing the information imbalance and misalignment between features while fusing. Libra RCNN [38] integrates features from each resolution to balance the information.

To construct more discriminative multiscale features, CEM enhanced the multiscale features before fusion by enriching the scaled information. A pixel-level information filtering strategy is applied in IJM during the fusion process, which filters and further fuses the discriminative information of the features. To shorten the information gap within features and balance semantic and spatial information, this study proposed SSM to aggregate deep and shallow features and realize local-to-global information interaction.

C. Lightweight Network Design

On-orbit object detection models are engineered to substantially decrease the volume of transmitted data, thereby minimizing latency and demonstrating. However, the on-orbit deployment is inherently restricted by stringent constraints related to power consumption, computational resources, and storage capacity. In this context, the implementation of a lightweight design becomes a pivotal factor in determining the overall performance of the model. The lightweight methods are mainly classified into four primary categories: weight quantization, knowledge distillation, network pruning, and convolutional layer replacement.

A more efficient quantization scheme is proposed that uses integers instead of floating points for inference [64]. Masked generative distillation (MGD) [65] masks the random pixels of the student features and forces the student features to generate the complete features of the teacher through a simple block. Differentiable Markov channel pruning (DMCP) [66] removes redundant or unimportant weights, neurons, or layers to achieve model compression and faster computational efficiency. A novel channel-level sparsity scheme compresses CNNs by reducing model size, memory usage, and computations while maintaining accuracy without specialized hardware or major training overhead [67]. Filter pruning removes low-impact CNN filters to reduce computation costs without adding a library of linear operations [68]. The automatic channel pruning (ACP) method [69] stores multiple pruned structures and then iteratively searches by particle swarm optimization to find the optimal compressed structure.

However, the lightweight methods mentioned above add additional modules to the model, which have limited effectiveness. The method of replacing the original convolutional layers with grouped convolutions or depth-separable convolutions is a much more prevalent and intuitive lightweight strategy. Xception [70] successfully improved model accuracy by replacing ordinary convolution with depth-separable convolution. MobileNets [71] used depth-separable convolution to build a lightweight deep neural network and introduced two hyperparameters, which enabled the selection of an appropriately sized model based on the constraints of the problem.

In this study, grouped convolution and depth-wise convolution are strategically utilized to reduce the parameters and FLOPs. Thanks to the efficient parameter utilization enabled by the proposed model architecture, the lightweight model can preserve its perceptual performance without degradation, while substantially decreasing its reliance on computational resources and power consumption.

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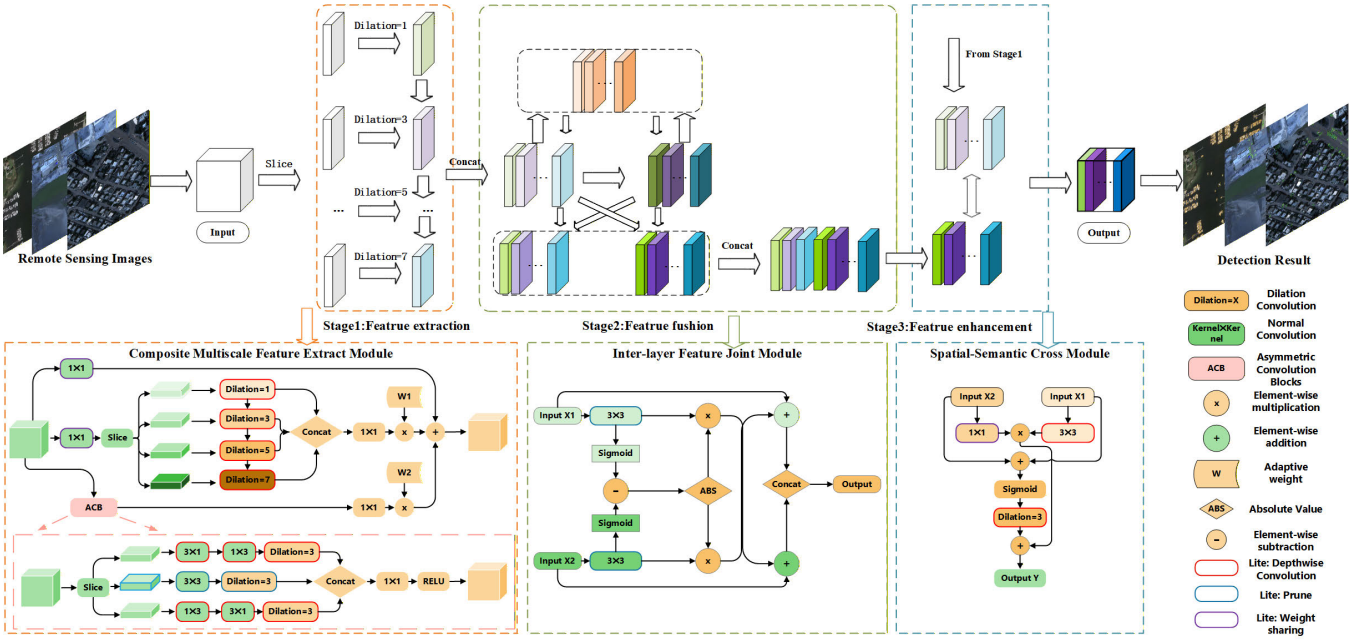


Fig. 3. Whole structure of MRN and MRN_lite. We used the CEM, IJM, and SSM modules in the multiscale feature extraction, fusion, and enhancement stages, respectively. Furthermore, MRN_lite is a lightweight version of MRN, and the changes are marked in the figure, including convolution layer replacement (red border), weight sharing (purple border), and pruning (blue border).

III. Proposed Method

A. Overview

To enhance the representation capability of features, as shown in Fig. 3, we propose three novel modules in MRN for reconstructing multiscale features. First, the CEM uses a multibranch parallel structure to extract features with different receptive fields, reconstructing multiscale features from single-resolution features. Next, the IJM filters the feature content, which reduces the proportion of irrelevant information during feature fusion. The SSM complements the spatial texture information of shallow features with the semantic features of higher level features, preventing a decline caused by information imbalance. In lightweight design, we are pruning, sharing weights, and alternating convolutional layers, which achieves the optimal balance between efficiency and cost. To validate the effectiveness of three modules and select the most balanced benchmark framework, we deploy the modules on YOLOv5 [39] and YOLOv11 [41] and evaluate the performance on the AITOD dataset. We finally chose YOLOv5 with fewer parameters as the benchmark model for MRN.

B. Composite Multiscale Feature Extraction Module

The richness of scale features is critical to the detection performance of small objects. In previous work, multiscale features are combined by different levels of feature maps, each of which only corresponds to a single scale. That results in multiscale features that contain less scale information. Thus, a composite multiscale feature enhancement module (FEM) is proposed for extracting multiscale features directly, rather than forming them later.

CEM is capable of aggregating several scales of information in a single feature map. In SOTA generic object detectors [56], [72], [73], multibranch convolution is used to avoid the single feature scale caused by sampling at the same center. However, branches are isolated from one another. Thus, we add an information flow channel between the branches to connect, and the output features of the previous branch will be delivered to the next branch for calculation. In addition, unlike the ordinary convolutions used in previous work [58], [60], dilated convolution is heavily used in CEM, which captures a larger range of contextual information. The formula of the atrous convolution block is written as

$$M_c^i = \text{Chunk}\,(M) \tag{1}$$

$$M_{\text{ms}}^i = \begin{cases} \text{Diconv}_{3\times3}^{d=1}\left(M_{\text{pre}}^i\right), & i = 1 \\ \text{Diconv}_{3\times3}^{d=2i-1}\left(M_{\text{pre}}^i \oplus M_{\text{ms}}^{i-1}\right), & i > 1 \end{cases} \tag{2}$$

$$\mathcal{F}(M) = \text{Concat}\left(M_{\text{ms}}^1, M_{\text{ms}}^2, \ldots, M_{\text{ms}}^n\right) \in \mathbb{R}^{(C_1+C_2+\cdots+C_n)\times H\times W}. \tag{3}$$

Concat denotes the splicing operation, and $\text{Diconv}_{3\times3}^{d=1}$ means a 3×3 atrous convolution with a dilation rate of 1. Chunk denotes a chunking operation along the channel, where the input features M are equally partitioned into n sets of M_c^i, where n is set to 4. M_{ms}^i is the output of the ith block of M_c^i after atrous convolution.

Furthermore, due to the diverse aspect ratios of the instances, we use asymmetric convolution to extract nonsquare features. The asymmetric convolution block also uses a multibranch convolution structure, which differs from the atrous convolution block in that it does not have information flow channels. Two branches are using a combination of 1×3 and 3×1 convolution kernels, and another branch with a 3×3 convolution kernel to avoid square instance features from disappearing. The formula of the asymmetric convolution

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block is expressed as

$$M_b^1 = \text{Diconv}_{3\times3}^{d=3}\left(\text{Asy}_{3\times1}\left(\text{Asy}_{1\times3}\left(M_c^1\right)\right)\right) \quad (4)$$

$$M_b^2 = \text{Diconv}_{3\times3}^{d=3}\left(\text{Asy}_{1\times3}\left(\text{Asy}_{3\times1}\left(M_c^2\right)\right)\right) \quad (5)$$

$$M_b^3 = \text{Diconv}_{3\times3}^{d=3}\left(\text{Conv}_{3\times3}\left(M_c^3\right)\right) \quad (6)$$

$$\mathcal{N}(M) = \text{Concat}\left(M_b^1, M_b^2, M_b^3\right). \quad (7)$$

M_c is defined by (1), and the number of groupings is set to 3 here. M_b^i is the output of the ith branch, and $\text{Asy}_{3\times1}$ represents asymmetric convolution with a 3×1 kernel size.

As shown in Fig. 3, the whole structure of CEM is composed of three main parts: atrous convolution block, asymmetric convolution block, and residual concatenation. The formula is expressed as follows:

$$Y = \mathcal{F}(M) \oplus \mathcal{N}(M) \oplus \text{Res}(M) \quad (8)$$

where Res denotes for the standard 1×1 convolution operation to construct residual concatenation.

C. Interlayer Feature Joint Module

Many experiments [54], [74], have demonstrated that deep and shallow features are complementary, and the accuracy of detection can be effectively improved by fusing them. However, previous works simply fuse features by splicing and adding, which do not deeply mine the discriminative content of features. These fusion methods weight the input features equally, resulting in mediocre performance. Despite the presence of FPN, the feature representation is still not rich enough. The bilinear channel fusion network (BCFN) is proposed to enhance the proposed feature representation [75]. The motivation stems from the fact that high-level features contain more semantic information, while low-level features are more likely to respond to local textures and patterns. Our motivation is similar; the semantic information of features is enhanced as the network deepens, while the detailed information weakens. IJM can capture changes in semantic and spatial information based on the gaps between neighboring features, and increase the intensity of changes during the process of fusing multiscale features. Therefore, we proposed IJM to fuse features, which filters out useful information by reweighting.

The difference between feature maps is a concentrated expression of the spatial and semantic information gap. IJM realizes multiscale feature fusion by exploiting the disparity of spatial and semantic information between features of different depths. Specifically, the gated mechanism is utilized to extract the discrepancy matrix between features. Then, we perform pixel-level reweighting of features according to the matrix. The discrepancy matrix M_{dis} is calculated by the following formula:

$$M_{\text{dis}} = \text{abs}\left(M_{\text{sig}}^d - M_{\text{sig}}^s\right) \quad (9)$$

where M_{sig} is the output of the input M_{in} after 3×3 convolution and Sigmoid activation function. M^d and M^s represent features from deep and shallow layers, respectively.

As shown in Fig. 3, the difference matrix M_{dis} will be used as the basis for reweighting features. IJM uses the matrix to guide the pixel-level fusion of interlayer features. The specific formula is as follows:

$$M_r^s = M_{\text{in}}^s + M_{\text{sig}}^d * M_{\text{dis}} \quad (10)$$

$$M_r^d = M_{\text{in}}^d + M_{\text{sig}}^s * M_{\text{dis}} \quad (11)$$

$$M_I = \text{Concat}\left(M_r^d, M_r^s\right). \quad (12)$$

M_r is the reweight result. M_I is composed of the reweighted deep and shallow features. IJM can reduce the gap between spatial information and semantic information of different depth feature maps and transfer each other's advantageous information.

D. Spatial-Semantic Information Cross Module

As the depth of the network deepens, the spatial-semantic information contained within high-level features becomes increasingly unbalanced, leading to the loss of detail. Shallow features are rich in texture and edge information but lack contextual information, while deep features are the opposite. To alleviate this imbalance, for each feature point pair (corresponding points in two features), we propose SSM to extract local–global information and aggregate it.

As shown in Fig. 3, SSM uses two different calculation methods to achieve local and global information interaction. Local information interaction makes full use of the correspondence between points to fuse features from different layers. And global information interaction sends features into a dilated convolution to extract information from a larger receptive field. The specific formula is shown as follows:

$$M_l = M_l^d * M_l^s \quad (13)$$

$$M_g = \text{Diconv}_{3\times3}^{d=2}\left(\text{Sig}\left(M_{\text{in}}^d + M_{\text{in}}^s\right)\right) \quad (14)$$

$$M_s = M_l + M_g. \quad (15)$$

Sig is the Sigmoid activation function. M_l and M_g are the feature maps of local and global information interaction, respectively. M_s is the output of SSM.

E. Lightweight Network

In the lightweight design of the model, how to avoid the loss of accuracy as much as possible and significantly reduce the cost is a problem worth exploring. In this article, we obtain MRN_lite by lightweighting MRN through pruning, depthwise convolution, and weight sharing.

The layer pruning algorithm [66] can significantly reduce the computational cost of the model by removing the convolutional layers that have less impact on the accuracy. In CEM, we believe that the features of square instances can be sufficiently characterized from the dilated convolutional block, so we eliminate the 3×3 convolutional branches from the asymmetric convolutional block. In addition, we remove two convolutional layers in IJM to further reduce the computational complexity of the model. After that, IJM will not use any convolutional operation to fuse features and will not increase any computational pressure on the device.

The method of group convolution strategy reduces the computational complexity of the model and reduces the number of model parameters. Without changing the sensory field range

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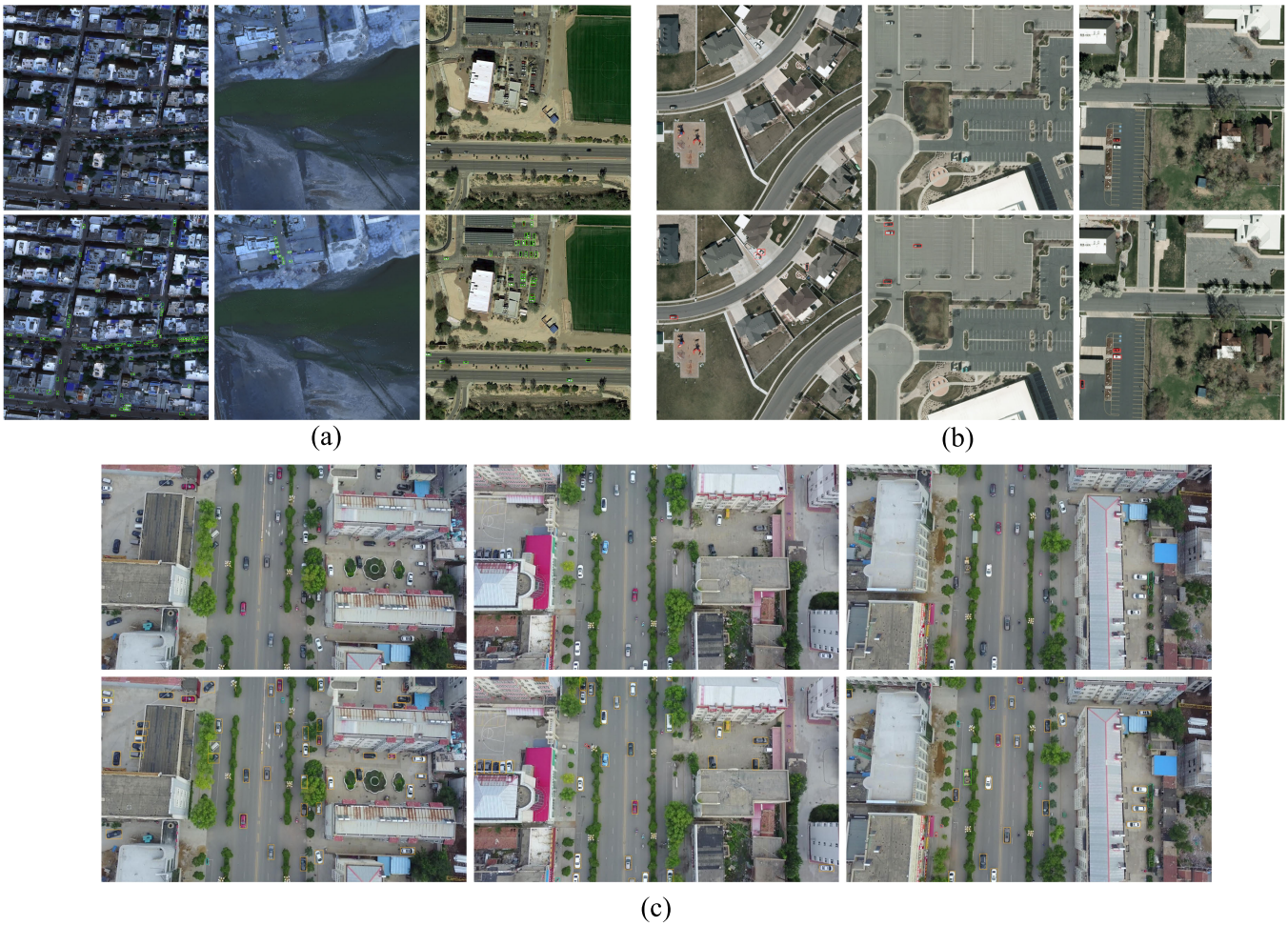


Fig. 4. Detection results of MRN_lite on three datasets. (a) AITOD. (b) VEDAI-512. (c) Visdrone2019.

of the convolutional layers, we use a special group convolution strategy (depth-wise convolution) for all the dilated convolutions in CEM and SSM, which maintains the model's sensitivity to multiple scales while drastically decreasing the number of floating-point operations of the model. In addition, we use shared weights to improve the utilization of the convolutional layers in SSM.

Fig. 3 shows three lightweight modules of MRN_lite. By pruning, using depth-wise convolution, and weight sharing, we achieve a lightweight design of the model. We first pruned the unimportant convolutional layers, then adopted a depth-by-depth convolution strategy to reduce the number of floating-point computations of the convolutional layers, and finally reused some of the convolutional layers to achieve weight sharing. Compared with the original version, the number of parameters decreased by 52%, from 14.2 to 6.7, and the number of floating points decreased by 50%, from 105.5 to 52.0. However, the performance of MRN_lite is very close to that of MRN on the three datasets.

IV. Results

A. Datasets and Evaluation Indicators

In this section, we validate the effectiveness of the proposed method on three widely used datasets: AITOD [76], VisDrone2019 [77], and VEDAI [78], which have different proportions of tiny objects, as shown in Fig. 2. We conduct comparison experiments on each of these three datasets, and the results show that MRN is insensitive to data distribution and maintains robustness to small objects. Furthermore, we will conduct ablation experiments on the AITOD dataset, which has a higher proportion of small objects, and evaluate the contribution of each module in detail. Fig. 4 demonstrates the detection results of MRN_lite on three datasets.

1) AITOD: AITOD consists of a training set of 11 214 images, a validation set of 2804 images, and a test set of 14 018 images. In total, it contains a dataset of 28 036 aerial images, providing 700 621 object instances for tiny object detection in aerial images. The average size of the object instances is about 12.8 pixels, where the maximum size of the instances does not exceed 64 pixels, and 86% of the instances are smaller than 16 pixels, which is much smaller than other datasets. AITOD contains eight categories, such as airplanes, bridges, storage tanks, etc., and is often used as a benchmark dataset for tiny object detection. To ensure the comparability of the experiments, as in other papers (support), we use the training and validation sets (totaling 14 018 images) for training and evaluate the performance of the model on the officially provided test set. In the comparison experiments, we

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evaluated our results according to the official evaluation tool [76], using APvt, APt, APs, and APm as evaluation metrics, which denote very tiny (less than 2×2), small (8×8 to 16×16 pixels), small (16×16 to 32×32), and medium (32×32 to 64×64 pixels) scaled objects of the AP.

2) VEDAI: A VEDAI dataset, which can be used as a tool for benchmarking automatic object recognition algorithms in an unconstrained environment. The database contains vehicles that, in addition to being small in size, exhibit varying variability, such as multiple orientations, illumination/shadow variations, specular reflections, or occlusions. Using satellite images from the AGRC in Utah and cropping them into smaller images, the database contains 1246 aerial images, and the dataset has large-scale color and infrared images at 1024×1024 resolution and small-scale color and infrared images at 512×512 resolution. It is worth noting that the small-scale color or infrared images are extracted from the corresponding large-scale images, except that they are scaled down to 512×512 to make the objects tiny. In addition, we performed experiments only on visible light images while disregarding categories with fewer than 50 instances. The number of detected categories is consistent with YOLO-fine [79] and SuperYOLO [80]. We will calculate the average precision in our comparison experiments and additionally evaluate tiny objects with scales smaller than 32×32, so the official YOLO metrics: mAP50, mAP50:95, and the official COCO metrics: APs will be used.

3) Visdrone: This dataset contains 10 209 drone images from different cities, different scenes, and different times of the day. The dataset is divided into three subsets: training set, validation set, and test set, with the ratio of the number of images between the subsets being 6471:548:1610. The dataset contains ten categories, such as pedestrians, people, buses, cars, etc., and the resolution of the images ranges from 480×360 to 2000×1500. In addition, these images were taken at different heights, with a large disparity of scales between the same type of objects. Same as previous work, since the evaluation server is down and it is not possible to evaluate the method in the test set, we use 548 images from the validation set for testing and 6471 images for training. COCO's official AP, AP50, and AP75 metrics will be used as our evaluation metrics in the comparison experiment.

B. Model Training Details

All experiments were performed on a workstation with NVIDIA 3090 GPUs. A stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005 was used to learn the parameters. At the same time, we introduce the normalized Wasserstein distance loss (NWDLoss) [14] as a complement to the edge loss. NWDLoss can effectively reduce the sensitivity of IOU metrics to tiny objects, and it is used together with CIOU to calculate the value of the object bounding box loss, with adaptive weights set to 0.5 for both. Experiments on the AITOD dataset set the epoch to 300, do not load the official pretraining weights, and set the image size to 640×640. Provided by the VEDAI dataset are two scale sizes of visible images: 512×512 and 1024×1024. We choose the lower resolution and tiny object 512×512 image as our dataset for evaluating the performance of the model, and the model is set to the same image size as the AITOD dataset to keep it consistent. As for the VisDrone dataset, to be able to compare and analyze with other SOTA experiments, we set the image size of the model to 1333×800, which is consistent with the EFC [54] method.

C. Comparative Experiments

MRN and MRN_lite are evaluated on three datasets: AITOD, VEDAI512, and Visdrone2019. Compared with other methods. Fig. 5 shows the detection results of MRN_lite compared with other methods.

1) Experimental Results on AITOD: As shown in Fig. 2, AITOD contains a high proportion of tiny objects with uniform distribution, which demonstrates the model's detection performance better. In order to reflect the model's ability to detect tiny objects, we adopt the evaluation metrics provided officially and use APvt, APt, APs, and APm to evaluate the model's ability to recognize and localize objects of different sizes. In comparative experiments, the performance improvement of nwdloss in small object detection is significant. We simply added this loss function to the official YOLOv5 model, and its performance surpassed most models developed in recent years. Therefore, to demonstrate the performance of the MRN and MRN_lite models, we added NWDLoss to the underlying architecture and included the results as part of the comparative experiments, thereby minimizing the influence of NWDLoss.

As shown in Table I, the MRN model improves by 3.5 percentage points compared to the baseline model that also uses the NWDLoss function, and compared to other SOTA models, the MRN achieves the lead in several metrics such as AP50:95, AP50, AP75, APvt, Apt, Aps, and APm. The lightweight version of MRN_lite outperforms MRN in the main evaluation metric AP, proving that the algorithm design of MRN_lite not only reduces the hardware threshold but also improves the detection capability.

It is worth noting that, in pursuit of higher detection accuracy, we also deployed MRN_lite on YOLOv11. Compared to the V5 version, MRN_lite (v11M) achieved outstanding detection performance, particularly improving AP and AP75 by 5.2 and 9.5 percentage points, respectively. However, the number of parameters and floating-point operations in the v11 version increased to 6.8 M and 140.7 G, respectively. For lightweight purposes, we ultimately chose V5 as our infrastructure and applied it to other datasets for comparative experiments.

2) Experimental Results on VEDAI: The officially provided 512×512 image is downsampled from a 1024×1024 image, and there is no difference in the content of the object contained in the two, but the object has fewer pixel units, which makes it more difficult to detect. We divide the training set and the test set according to the official method. To ensure the uniformity of the dataset, for CMAFF [94] and SuperYOLO [80], we use their training results on RGB images only for comparison. As shown in Table II, MRN and MRN_lite are increased by 3.4 and 3.0 percentage points on mAP50 compared with

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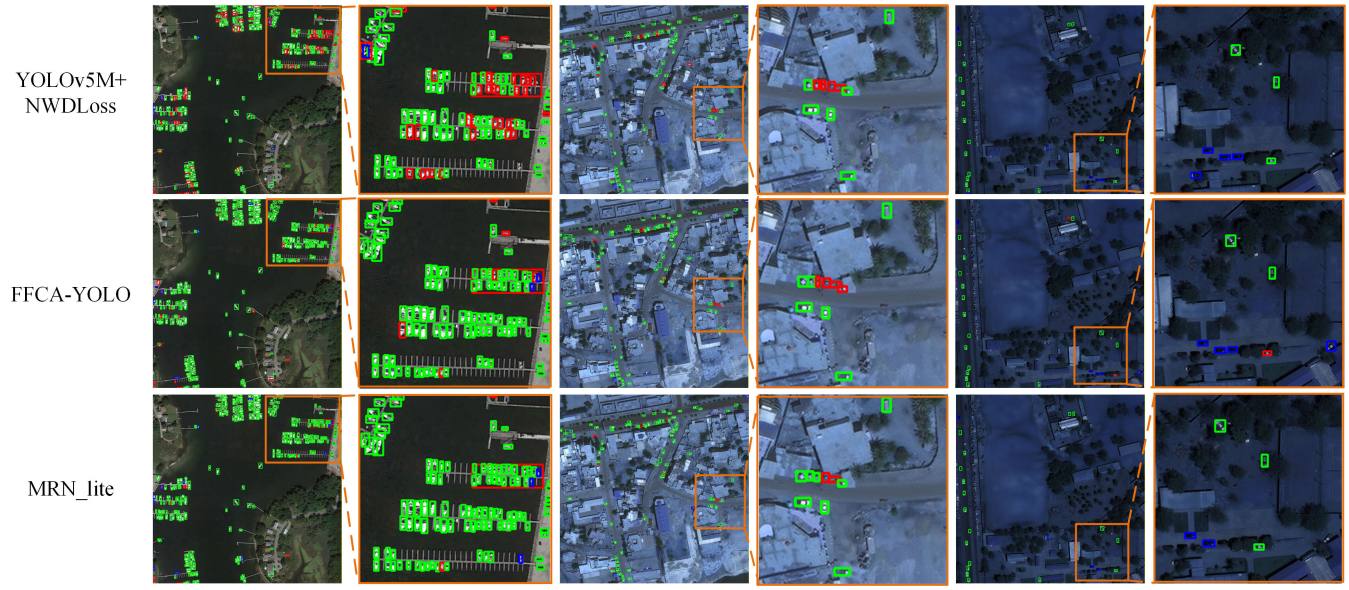


Fig. 5. Visualization of detection results on the AITOD dataset. The orange box is a local magnified figure of the heatmap. The green, blue, and red boxes denote true positive (TP), false positive (FP), and false negative (FN) predictions, respectively.

TABLE I
COMPARISONS EXPERIMENTS FOR MRN AND MRN_LITE IN AITOD * INDICATES THAT THE MODEL USES NWDLOSS

Model	Venue	AP	AP_{50}	AP_{75}	APvt	APt	APs	APm	Param/M	FLOPs/G
Faster RCNN [1]	TPAM12017	11.1%	26.3%	7.6%	0.0%	7.2%	23.3%	33.6%	236.3	289.3
Cascade RCNN [2]	CVPR2018	13.8%	30.8%	10.5%	0.0%	10.5%	25.5%	36.6%	319.4	146.6
DETR [5]	ECCV2020	2.7%	10.3%	0.7%	0.7%	2.1%	3.0%	12.4%	41.0	225.0
ATSS [81]	CVPR2021	12.8%	30.6%	8.5%	1.9%	11.6%	19.5%	29.2%	244.5	-
CenterNetV2 [82]	Arxiv2021	14.1%	35.7%	8.5%	2.9%	12.7%	19.7%	29.5%	44.9	206.0
DetectoRS [49]	CVPR2021	14.8%	32.8%	11.4%	0.0%	10.8%	28.3%	38.0%	540.1	246.0
Deformable-DETR [83]	ICLR2021	17.0%	45.9%	8.8%	7.2%	17.1%	22.7%	28.2%	40.0	196.0
DAB-Deformable-DETR [84]	ICLR2022	16.5%	42.6%	9.9%	7.9%	15.2%	23.8%	31.9%	44.0	256.0
FSANet [85]	TGRS2022	20.3%	48.1%	14.0%	6.3%	19.0%	26.8%	36.7%	31.9	125.2
NWD-RKA [86]	ISPRS2022	23.4%	53.5%	16.8%	8.7%	23.8%	28.5%	36.0%	540.1	246.0
YOLOv8m [40]		22.9%	52.5%	16.5%	7.1%	22.4%	29.7%	37.1%	25.8	78.7
KLDet [87]	TGRS2024	19.6%	-	-	8.4%	20.6%	22.7%	26.4%	32.0	-
RFLA [88]	ECCV2022	24.8%	55.2%	18.5%	9.3%	24.8%	30.3%	38.2%	-	-
ORFENet [89]	TGRS2024	24.8%	55.4%	18.2%	9.7%	24.4%	28.7%	35.1%	32.7	373.3
HANet [90]	TCSVT2023	22.1%	53.7%	14.4%	10.9%	22.2%	27.3%	36.8%	26.4	-
MENet-Swin-T [91]	TGRS2024	23.2%	56.2%	15.0%	9.7%	23.9%	25.3%	34.4%	41.9	148.3
DINO-Deformab1e-DETR [84]	ICLR2023	23.2%	56.6%	15.4%	9.9%	23.1%	29.3%	37.6%	44.0	279.0
BRSTD [92]	TGRS2024	23.6%	54.2%	16.6%	10.4%	22.8%	30.3%	37.6%	1.8	64.3
BRSTD_l [92]	TGRS2024	26.1%	58.0%	19.5%	11.6%	26.0%	32.3%	38.5%	6.3	220.2
FFCA-YOLO* [56]	TGRS2024	29.7%	68.0%	20.2%	12.0%	30.8%	34.5%	35.0%	7.1	51.3
YOLOv11m* [41]		30.4%	62.9%	24.4%	8.0%	29.2%	39.8%	44.8%	20.0	68.2
YOLOv5m* [39]		27.6%	64.3%	19.3%	9.0%	28.6%	33.7%	32.8%	20.8	47.9
MRN*		**31.1%**	**69.3%**	**22.1%**	**13.0%**	**32.5%**	**35.7%**	**35.9%**	**14.2**	**106.0**
MRN_lite*		**31.4%**	**69.1%**	**23.0%**	**12.2%**	**33.6%**	**36.2%**	**35.8%**	**6.8**	**52.0**
MRN_lite*(v11m)		**36.6%**	**70.8%**	**32.8%**	**12.9%**	**36.9%**	**44.5%**	**44.6%**	**11.0**	**140.7**

the best performance: FFCA-YOLO [56]. This highlights the superiority of our method for small object detection. However, the performance of MRN and MRN_lite is similar to that of other methods at mAP50:95 and mAPs. We believe this is mainly due to the small size of the VEDAI test set, and the accuracy of each image can have a significant impact on the final test results. On the other hand, due to the unconcentrated size distribution of the instances, the optimization process of the model is seriously disturbed by scale changes, as shown in Fig. 2.

3) Experimental Results on Visdrone: The VISDRONE dataset is derived from UAVs, with a wide range of shooting angles and obvious variations in the size of similar objects, which can verify the robustness of the model. As shown in Table III, compared to EFC combined with GFL, MRN reduces parameters and FLOPS by more than 77.8% and 64.4% and improves by 0.3% at AP50:95 and 0.6% at AP75, respectively. In the meantime, the parameter and FLOPS of MRN_lite are reduced by more than 89.1% and 82.9%, and AP50:95, AP50, and AP75 are only decreased by 0.5%, 1.2%,

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TABLE II
COMPARISONS EXPERIMENTS FOR MRN AND MRN_LITE IN VEDAI

Methods	Car	Pickup	Camping	Truck	Other	Tractor	Boat	Van	mAP_{50}	$mAP_{50:95}$	mAPs
Lightweight CNN [93]	74.7%	56.7%	56.7%	36.1%	26.9%	56.7%	22.7%	36.1%	52.6%	-	-
YOLO-fine [79]	76.7%	74.3%	64.7%	63.4%	45.0%	78.1%	70.0%	77.9%	68.1%	-	-
SuperYOLO(RGB) [80]	90.3%	82.6%	76.6%	68.5%	53.8%	79.4%	58.0%	70.3%	72.4%	-	-
CMAFF(RGB) [94]	91.7%	85.9%	75.1%	78.3%	33.3%	81.2%	71.8%	62.2%	74.3%	-	-
TPH-YOLO [95]	84.0%	76.4%	60.7%	62.9%	38.3%	63.5%	23.7%	57.3%	58.4%	33.8%	34.5%
YOLOv5m [39]	86.6%	78.7%	72.4%	60.7%	71.7%	79.7%	56.0%	73.6%	72.3%	41.0%	39.9%
YOLOv8m [40]	85.9%	81.0%	61.7%	83.9%	56.0%	78.3%	42.6%	59.5%	68.6%	40.8%	40.1%
FFCA-YOLO [56]	89.6%	85.7%	78.7%	85.7%	48.6%	81.8%	61.5%	67.0%	74.8%	44.8%	44.6%
L-FFCA-YOLO [56]	91.3%	85.5%	72.8%	79.7%	47.3%	79.0%	56.1%	73.9%	73.3%	44.7%	44.5%
MRN	**90.8%**	**84.2%**	**87.3%**	**68.0%**	**68.6%**	**85.0%**	**63.0%**	**63.2%**	**76.3%**	**43.8%**	**42.3%**
MRN_lite	**92.2%**	**87.7%**	**83.9%**	**73.9%**	**68.1%**	**80.4%**	**54.9%**	**72.1%**	**76.7%**	**45.6%**	**44.2%**

TABLE III
COMPARISONS EXPERIMENTS FOR MRN AND MRN_LITE IN VISDRONE2019

Method	AP	AP_{50}	AP_{75}	FLOPs/G	Params/M
YOLOv3-SPP3 [96]	26.4%	-	-	284. 1	63.9
ClusDet [97]	28.4%	56.2%	31.6%	-	-
Improved CascadeNet [43]	28.8%	47.1%	29.3%	-	-
DMNet [98]	29.4%	49.3%	30.6%	-	-
YOLC [99]	29.7%	52.4%	29.2%	-	-
Faster R-CNN [1]	24.8%	43.6%	25.0%	322.3	30.0
RetinaNet [42]	20.2%	36.9%	19.5%	586.8	42.7
QueryDet [62]	19.6%	35.7%	19.0%	-	-
QueryDet-CSQ [62]	19.3%	35.0%	18.9%	-	-
CEASC [100]	20.8%	35.0%	27.7%	201.96	-
GFL [101]	28.4%	50.0%	27.8%	525.0	42.5
MobileNet V2 [102]	28.5%	50.2%	28.1%	491.5	-
ShuffleNet V2 [103]	26.2%	46.6%	25.7%	488.9	-
CEASC [100]	28.7%	50.7%	28.4%	150.2	-
RetinaNet+EFC [54]	23.3%	39.2%	23.9%	482.3	40.2
Faster R-CNN+EFC [54]	25.6%	44.8%	25.7%	301.4	28.5
GFL+EFC [54]	30.1%	52.1%	29.8%	477.6	40.0
MRN	**30.4%**	**51.8%**	**30.4%**	**106.0**	**14.2**
MRN_lite	**29.6%**	**50.9%**	**29.3%**	**52.0**	**6.8**

TABLE IV
ABLATION EXPERIMENTS FOR MRN_LITE IN AITOD

CEM	IJM	SSM	Precision	Recall	mAP_{50}	$mAP_{50:95}$	FLOPs/G	Param/M
✗	✗	✗	0.708	0.614	0.623	0.277	45.1	6.517
✓	✗	✗	0.731	0.609	0.626	0.275	49.0	6.664
✓	✓	✗	0.717	0.623	0.633	0.281	49.0	6.665
✓	✓	✓	0.726	0.621	0.641	0.283	52.0	6.873
✓	✗	✓	0.718	0.620	0.632	0.280	52.0	6.872
✗	✓	✓	0.728	0.618	0.630	0.283	48.0	6.725
✗	✗	✓	0.713	0.623	0.628	0.281	48.0	6.724
✗	✓	✗	0.721	0.620	0.631	0.277	45.1	6.517

and 0.5%. Therefore, our proposed method not only achieves a better balance between accuracy, parameters, and FLOPS but also is insensitive to the object size and photographic viewpoints, which can effectively deal with UAV object detection under complex photographic angles and altitudes.

D. Results of Ablation Experiments

To analyze the effect of each module on the MRN_lite network, we applied CEM, IJM, and SSM to the baseline in stages to test their performance. In the ablation experiments, we used YOLO's official precision, recall, mAP50, and mAP50:95 as the evaluation indices. Table IV shows the experimental results of different module combinations.

In addition, to demonstrate the role of each module, we visualize the feature map of each module, as shown in Fig. 6. Each highlighted red pixel represents the model's prediction for the interested objects. The CEM extracts information about the object and its surrounding context. The extracted features are mixed at the preliminary stage, including the object features

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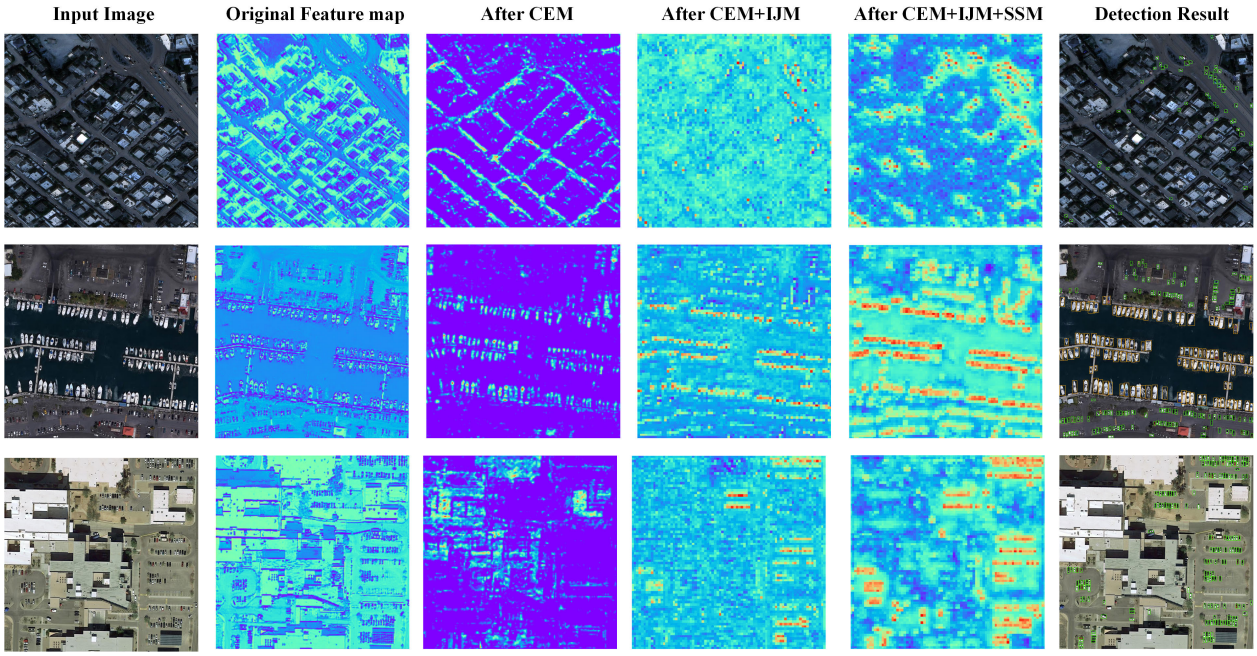


Fig. 6. Influence of CEM, IJM, and SSM on multiscale feature. The red represents that the model pays more attention to that area.

TABLE V
COMPARISON EXPERIMENTS FOR CEM IN AITOD

Method	Precision	Recall	mAP_{50}	$mAP_{50:95}$	FLOPs	Param
FEM [56]	0.701	0.621	0.625	0.277	52.3	6.918
PKI Module [104]	0.697	0.572	0.608	0.266	55.1	7.038
PKI Block [104]	0.703	0.610	0.623	0.272	61.0	7.322
CEM (ours)	**0.726**	**0.621**	**0.641**	**0.283**	**52.0**	**6.873**

TABLE VI
COMPARISON EXPERIMENTS FOR IJM IN AITOD

Method	Precision	Recall	mAP_{50}	$mAP_{50:95}$	FLOPs	Param
FFM [56]	0.692	0.593	0.609	0.260	52.5	6.900
ECA [105]	0.704	0.613	0.619	0.269	52.0	6.862
EFC [54]	0.701	0.610	0.621	0.271	56.8	7.149
IJM (ours)	**0.726**	**0.621**	**0.641**	**0.283**	**52.0**	**6.873**

(e.g., ship features), highly related background information (e.g., road features), and unrelated information (e.g., building features). After enhancement by the IJM, the extent of the red regions increases, capturing the approximate location of the object. However, compared with SSM, the red regions in IJM features are sparser and have blurred boundaries, reflecting confusion in dense object scenarios (such as closely situated ships) and a degradation of spatial details. This often causes adjacent objects to merge into a single oversized activation area, thereby limiting the ability to distinguish objects. Through the spatial supplementation provided by SSM, the heatmap contours become clearer and more precise. This spatial refinement reduces the risk of misidentification (such as merging or missing adjacent objects), allowing attention to focus on precise object boundaries, which plays a key supporting role in downstream localization tasks.

1) Ablation Experiments of CEM: CEM can improve all evaluation metrics on the AITOD dataset, especially in terms of accuracy (from 0.708 to 0.731). The results show that CEM extracts sufficiently rich scale features, allowing the model to recognize and distinguish the object from the background from different scale perspectives. Compared with the baseline, the CEM module alone can improve the performance of the network by only increasing the number of parameters by 0.147 M and the number of floating-point computations by 3.9 G. In Table V, we replaced CEM with modules such as the FEM [56], poly kernel inception block (PKI Block), and PKI module [104]. FEM adopts a multibranch convolution structure, but it lacks information flow channels and cannot fully utilize the connections between branches. And CEM introduces information flow channels and preserves the complete feature representation of each branch, resulting in the best performance.

To further verify the effect, we visualize the features before and after CEM. The red region indicates the region that the model pays more attention to. Thanks to the very rich feature scales extracted by CEM, MRN_lite links the contextual information of different scales in the process of recognizing the object and accurately filters the region where the object is most likely to appear.

2) Ablation Experiments of IJM: As shown in Table IV, IJM puts almost no burden on the network in terms of computational power and storage and effectively improves the precision (from 0.708 to 0.721) and mAP50 (from 0.623 to 0.631) of the network. A gating mechanism is adopted in IJM, which filters and enhances the useful information of the feature at the pixel level. As shown in Fig. 6, the model recognizes more objects (red area) after IJM, which indicates that the IJM module can further strengthen the network's ability to recognize objects. In addition, Table VI also shows that IJM has a significant advantage over other fusion strategies. Compared with enhanced interlayer feature

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TABLE VII
COMPARISON EXPERIMENTS FOR SSM IN AITOD

Method	Precision	Recall	mAP_{50}	$mAP_{50:95}$	FLOPs	Param
SCAM [56]	0.714	0.621	0.629	0.278	55.7	8.232
MGA [106]	0.702	0.615	0.624	0.274	73.0	9.866
CBAM [107]	0.699	0.623	0.632	0.280	56.4	9.101
SSM (ours)	**0.726**	**0.621**	**0.641**	**0.283**	**52.0**	**6.873**

correlation (EFC) [54], IJM is more lightweight and performs better on mAP50 and mAP50:95, improving by 2 and 1.2 percentage points, respectively.

IJM does not use any convolution operation in MRN_lite; thus, there is almost no increase in the number of parameters and FLOPs.

3) Ablation Experiments of SSM: To alleviate the imbalance of spatial and semantic information, we adopt SSM to cross-combine the deep feature and the shallow one. As shown in Fig. 6, compared with the feature map output from IJM, the red area of the feature map is deepened, and the contour is more obvious after SSM. This phenomenon indicates that SSM can effectively supplement spatial texture information and semantic information of the object and strengthen the model's ability to localize and distinguish dense objects. Table IV shows the comparison results, and it shows that SSM improves the performance of the network in terms of precision, recall, mAP50, and mAP50:95 evaluation metrics.

In Table VII, we introduce convolutional block attention module (CBAM) [107], multiorder gated aggregation (Moga) block [106], and spatial context aware module (SCAM) [56] methods for comparison. The experiment proves that the dual-branch information interaction method adopted by SSM supplements more spatial information.

V. CONCLUSION

In this work, we propose three innovative modules: CEM, IJM, and SSM, which reconstruct multiscale features in the extraction, fusion, and enhancement processes, respectively. The CEM module constructs multiscale features with rich semantics within a single feature map, providing input features with richer scale information for the feature fusion stage. IJM uses a gating mechanism to filter and strengthen the spatial information of shallow features and the semantic information of deeper features when fusing features from different levels to further enhance the discriminative properties of the features. SSM is focused on supplementing the spatial information of deep features and achieving a balance between spatial and semantic information. We integrate these three modules into the FPN to obtain the MRN network. At the same time, to reduce the model's requirements on the equipment, we adopt the depth-wise convolution and manual pruning to a lightweight MRN, and a lightweight version of the network, MRN_lite, is obtained. MRN_lite drastically reduces the number of parameters and FLOPs of the original network, which shows comparable performance to the original network on three small object datasets: AITOD, VEDAI (RGB), and Visdrone. MRN and MRN_lite both outperform the previous SOTA method on three datasets.

Improving the detection accuracy of small objects remains a challenging yet promising research focus. Our future work will further expand the sources of data, going beyond just visible light. By utilizing other spectral images, such as infrared and SAR images, the impact of extreme conditions like lighting and fog on the model's detection capability can be reduced. This also aligns with the reality of remote sensing platforms carrying diverse payloads. Therefore, we will next focus on studying cross-modal object detection methods, leveraging multisource data to increase the information content of small objects and enhance the model's detection performance and robustness.

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